

Bounded nilpotency step forces non-degeneracy in direct systems of Carnot groups

Candidate full solution to Problem P3 of arXiv:2101.03979

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Status. Candidate full solution, likely valid; human review needed. The answer to Problem P3 is affirmative.

Source problem

Moisala and Pasqualetto ask in Problem P3 [1, p. 3]:

The complete picture of the relation between nilpotency of the groups G_i and non-degeneracy of the direct system remains unclear; we record the following open problems.

P2) Can we remove the nilpotency assumption in the statement of Theorem 1.2?

P3) If $(\{G_i\}_{i \in I}, \{\varphi_{ij}\}_{i \leq j})$ is a direct system of infinite-dimensional Carnot groups such that the nilpotency steps of the groups G_i are uniformly bounded, is it non-degenerate?

In words: if a direct system of infinite-dimensional Carnot groups has a uniform bound on the nilpotency steps, must it be non-degenerate in the category of complete metric scalable groups (CMSGs)?

Proof intuition

The obstruction in the source example is that conjugation can send elements tending to the identity a definite distance away. In a group of fixed nilpotency class this cannot happen. A homogeneous component of weighted degree r can be recovered from an element by a bounded collection of dilated group words. Applying that collection to a commutator, and choosing the two dilation parameters asymmetrically, gives the scale-invariant estimate

$$\|\exp[U, V]\| \lesssim_s \|\exp U\|^{p/(p+q)} \|\exp V\|^{q/(p+q)} \quad (U \in V_p, V \in V_q).$$

Every nontrivial BCH term in $g^{-1}hg$ contains at least one homogeneous component of h and has total degree at most s . It therefore carries at least the power $\|h\|^{1/s}$. Crucially, all constants depend only on s , not on the dimension, structure constants, or homogeneous metric. This uniformity lets the estimate pass first through the finite-dimensional filtration of each G_i and then through the infimum pseudometric of the direct system.

Homogeneous norms and a uniform collection lemma

If G is a Carnot group with dilation δ , write

$$\|g\| = d(e, g).$$

Thus $\|uv\| \leq \|u\| + \|v\|$, $\|u^{-1}\| = \|u\|$, and $\|\delta_t u\| = |t|\|u\|$. No geodesic or sub-Finsler assumption is made.

We record the only nilpotent-collection input. It is important that its constant is uniform over the group and the norm.

Lemma 1 (bounded homogeneous collection). *For every $s \geq 1$ there is $A_s < \infty$ with the following property. Let $\mathfrak{g} = V_1 \oplus \cdots \oplus V_s$ be any graded nilpotent Lie algebra, let G be its simply connected group, and equip G with any homogeneous left-invariant distance. Every $z \in G$ has an ordered factorization*

$$z = z_1 z_2 \cdots z_s, \quad z_r \in \exp(V_r), \tag{1}$$

such that

$$\|z_r\| \leq A_s \|z\| \quad (1 \leq r \leq s). \quad (2)$$

The same assertion holds for a homogeneous pseudonorm by approximation.

Proof. This is the metric form of finite nilpotent collection. We prove the slightly stronger assertion that every z_r is a limit of words in dilated copies of z and their inverses whose *dilation length* (the sum of the absolute dilation parameters, with integer powers expanded) is bounded only in terms of s . Such a word W automatically satisfies

$$\|W(z)\| \leq \text{len}_\delta(W) \|z\|. \quad (3)$$

Proceed by induction on s . Class one is immediate. For the induction step, let $Z = \exp(V_s)$ and let $\pi : G \rightarrow G/Z$. By induction, every homogeneous factor $\bar{z}_r$ of $\pi(z)$, $r < s$, is the limit of words $W_{r,n}(\pi z)$ of dilation length at most A_{s-1} . Lift the same words to G . Equation (3) puts $W_{r,n}(z)$ in the fixed compact ball of radius $A_{s-1} \|z\|$ (homogeneous balls in a finite-dimensional graded group are compact). After taking a subsequence, it converges to an element

$$w_r = \exp(W_r + C_r), \quad W_r \in V_r, \text{quad} C_r \in V_s,$$

whose projection is $\bar{z}_r$. The top layer is central, so its error is peeled off by the exact two-dilate identity

$$\delta_2(\exp(W_r + W_s)) (\exp(W_r + W_s))^{-2^r} = \exp((2^s - 2^r)W_s), \quad r < s, \quad (4)$$

with $W_s = C_r$. After the dilation $\delta_{|2^s - 2^r|^{-1/s}}$, this gives $\exp C_r$ at norm at most

$$D_{r,s} \|w_r\|, \quad D_{r,s} = \frac{2 + 2^r}{|2^s - 2^r|^{1/s}}.$$

Consequently $z_r := w_r \exp(-C_r) \in \exp(V_r)$ and $\|z_r\| \leq (1 + D_{r,s}) A_{s-1} \|z\|$. Substitution of the lifted word sequences into (4) also preserves the stronger bounded-word assertion, with a new constant depending only on s .

Having obtained $z_1, \dots, z_{s-1}$, set

$$z_s = (z_1 \cdots z_{s-1})^{-1} z.$$

Its image under π is the identity, so $z_s \in Z = \exp(V_s)$, and subadditivity bounds it by $\|z\| + \sum_{r < s} \|z_r\|$. Taking the maximum of these finitely many recursive constants defines A_s . This proves both (1) and (2), with no dimension, structure constant, or metric equivalence constant. A homogeneous pseudonorm follows by adding an arbitrarily small genuine homogeneous norm and letting its coefficient tend to zero. $\square$

The uniform conjugation estimate

Lemma 2 (weighted commutator estimate). *If $U \in V_p$, $V \in V_q$, and $p + q \leq s$, then*

$$\|\exp[U, V]\| \leq B_s \|\exp U\|^{p/(p+q)} \|\exp V\|^{q/(p+q)}, \quad (5)$$

where B_s depends only on s .

Proof. Put $u = \exp U$, $v = \exp V$, $P = \|u\|$, and $Q = \|v\|$. For $a, b > 0$, the first nonzero homogeneous factor of $[\delta_a u, \delta_b v]$ is $\exp(a^p b^q [U, V])$. Lemma 1 and the commutator word give

$$(a^p b^q)^{1/(p+q)} \|\exp[U, V]\| \leq A_s \|\delta_a u, \delta_b v\| \leq 2A_s(aP + bQ).$$

When $P, Q > 0$, choose $a = P^{-1}$ and $b = Q^{-1}$. This gives (5) with $B_s = 4A_s$. The cases $P = 0$ or $Q = 0$ follow by approximation (and are trivial for a genuine norm). $\square$

Iteration of Lemma 2 gives the corresponding estimate for every homogeneous Lie monomial. If a monomial M has total weighted degree $d \leq s$, and its entries E_j have weights r_j , then

$$\|\exp M(E_1, \dots, E_m)\| \leq C_s \prod_j \|\exp E_j\|^{r_j/d}. \quad (6)$$

Rational coefficients in BCH are absorbed into C_s , since a scalar on V_d changes the homogeneous norm by its $1/d$ power. Finite sums of such monomials are collected by the same triangular word process as in Lemma 1; the correction commutators have higher weight and obey (6) again. Thus their exponential is bounded by a constant times the sum of the displayed monomial bounds.

Theorem 3 (uniform conjugation modulus). *For every $s \geq 1$ there is $C_s < \infty$ such that, in every Carnot group of step at most s with every homogeneous left-invariant distance,*

$$\|g^{-1}hg\| \leq C_s(1 + \|g\|)^s \|h\|^{1/s} \quad \text{whenever } \|h\| \leq 1. \quad (7)$$

The same estimate holds in every infinite-dimensional Carnot group that is algebraically nilpotent of class at most s .

Proof. For a finite-dimensional group, use Lemma 1 to write g and h as products of at most s pure homogeneous factors, each bounded by $A_s\|g\|$ or $A_s\|h\|$, respectively. Equivalently, apply the finite BCH formula to

$$\log(g^{-1}hg) = e^{-\text{ad}(\log g)} \log h$$

and then collect the result into homogeneous factors. Every nonconstant Lie monomial that occurs contains at least one homogeneous component coming from h ; all remaining entries come from g . If $d \leq s$ is its total weight and $q \geq 1$ is the total weight contributed by its h -entries, then (6) bounds that factor by

$$C_s(1 + \|g\|)^s \|h\|^{q/d} \leq C_s(1 + \|g\|)^s \|h\|^{1/s},$$

because $q/d \geq 1/s$ and $\|h\| \leq 1$. There are only finitely many universal BCH terms of weight at most s . Subadditivity proves (7).

Now let G be an infinite-dimensional Carnot group of class at most s and let (N_m) be a filtration by finite-dimensional Carnot subgroups. Every N_m has class at most s , so the estimate just proved has the same constant on all N_m . Approximate g, h by elements of the increasing union $\bigcup_m N_m$ and pass to the limit. Multiplication and inversion are continuous in the already-given CMSG G , so (7) follows on G . $\square$

For comparison, the classical fixed-group estimate has the same exponent $1/s$, but its stated constant depends on the particular group and metric; see Magnani [2, Lemma 3.6]. The uniformity in Theorem 3 is what is needed for a direct system.

Affirmative solution of P3

Let

$$\mathcal{G} = (\{G_i\}_{i \in I}, \{\varphi_{ij}\}_{i \leq j})$$

be a direct system as in P3, and suppose every G_i has nilpotency class at most s . Let (G', φ'_i) be its direct limit as scalable groups and let d' be the source's infimum pseudometric.

Theorem 4 (answer to Problem P3). *The direct system $\mathcal{G}$ is non-degenerate in the category of CMSGs. More precisely, for $x = \varphi'_i(x_i)$ and $M = d_i(e_i, x_i)$, right translation satisfies the uniform-at- x modulus*

$$d'(x, yx) \leq C_s(1 + M)^s d'(e', y)^{1/s} \quad \text{when } d'(e', y) \leq 1, \quad (8)$$

with the right side interpreted by an arbitrarily close upper approximation if the infimum defining d' is not attained.

Proof. Fix $x = \varphi'_i(x_i)$. Let $y \in G'$ and choose any $\eta > d'(e', y)$. By the definition of the infimum pseudometric, there are a stage j and $a_j, b_j \in G_j$ such that

$$\varphi'_j(a_j) = e', \quad \varphi'_j(b_j) = y, \quad d_j(a_j, b_j) < \eta.$$

Set $h_j = a_j^{-1}b_j$. Then $\varphi'_j(h_j) = y$ and, by left invariance, $\|h_j\|_j < \eta$.

Choose a common upper stage $k \geq i, j$ and put

$$x_k = \varphi_{ik}(x_i), \quad h_k = \varphi_{jk}(h_j).$$

The bonding maps are 1-Lipschitz, so $\|x_k\|_k \leq M$ and $\|h_k\|_k < \eta$. If $\eta \leq 1$, Theorem 3 gives

$$\begin{aligned} d'(x, yx) &\leq d_k(x_k, h_k x_k) = \|x_k^{-1} h_k x_k\|_k \\ &\leq C_s(1 + M)^s \eta^{1/s}. \end{aligned}$$

Letting $\eta \downarrow d'(e', y)$ proves (8). In particular, every right translation on (G', d') is continuous at the identity; this is condition (C2) of [1]. Notice that neither injectivity of the bonding maps nor attainment of the infimum was used.

It remains to check the two CMSG conditions rather than only (C2). By source Lemma 3.10, (C2) makes the metric quotient $G'/Z(d')$ a metric scalable group. It has class at most s : any $(s+1)$ -fold commutator has finitely many entries, hence all are represented at one common stage, where that commutator vanishes. Source Lemma 3.17 on this quotient therefore upgrades (C2) to Cauchy-continuity of inversion, which is condition (C3) for the system. Moreover the first layer generates the quotient as a topological scalable group by the argument in source equation (4.3); source Lemma 4.7 then gives the dilation condition (C1). Thus $\mathcal{G}$ is non-degenerate as a direct system of CMSGs. $\square$

Corollary 5. *Every direct system in P3 has a direct limit in the category of CMSGs. If the indexing set is countable, that limit is an infinite-dimensional Carnot group.*

Proof. Apply source Theorem 3.15 for existence. In the countable case apply source Theorem 4.8 (equivalently Theorem 1.2) to the non-degenerate system. $\square$

Verification, limitations, and novelty

Eight focused attempts were recorded. Counterexample attempts based on growing structure constants and changing homogeneous metrics fail precisely because the optimized commutator estimate forces the newly created layer to shrink. Classes two and three were collected explicitly; in class three, if $r = \exp(Y_2 + Z_3)$ then $\delta_2(r)r^{-4} = \exp(4Z_3)$, confirming the uniform layer extraction and the $1/3$ exponent. The filtration passage, noninjective bonding maps, near-minimizers for d' , and the source's separate conditions (C1)–(C3) were then audited.

The theorem does not address P1 or P2. It resolves exactly P3. The principal review point is Lemma 1: its finite collection constants must depend only on the truncation weight s . The free graded construction and central peeling identity make this dependence explicit and exclude dimension, structure constants, and metric equivalence constants.

A bounded novelty search through 12 August 2026 covered the run indexes, arXiv-id and exact-title searches, the quoted wording of P3, and searches for bounded-step non-degeneracy and homogeneous conjugation estimates. It found the source and its published version, plus the fixed-group estimate in [2], but no later paper claiming an answer to P3. Novelty confidence is moderate: the ingredients are standard nilpotent collection tools, while the uniform direct-system consequence appears not to have been recorded.

Human-review recommendation. Check the bounded-word proof of Lemma 1 first, then verify the near-minimizing-pair step in Theorem 4. Once those two points are accepted, the remaining implications are explicit source lemmas.

References

- [1] T. Moisala and E. Pasqualetto, *Direct limits of infinite-dimensional Carnot groups*, Math. Scand. 128 (2022), no. 2, 160–200; arXiv:2101.03979, doi:10.7146/math.scand.a-132062.
- [2] V. Magnani, *Towards differential calculus in stratified groups*, J. Aust. Math. Soc. 95 (2013), no. 1, 76–128; arXiv:math/0701322.